CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Startup businesses play a significant role in modern economic growth by introducing innovative products, services, and technologies. However, starting and managing a business requires strong decision-making skills, strategic planning, and an understanding of market dynamics. Many students and aspiring entrepreneurs learn business concepts mainly through theoretical knowledge, which may not provide practical exposure to real-world business challenges.

This project introduces an intelligent framework for modelling and simulating startup business processes through a game-based environment. The system allows users to experience the different stages involved in creating and managing a startup company. The simulation environment helps users understand how business decisions influence company growth, financial performance, and market success.

The proposed system is developed using the Unity Game Engine and integrates Artificial Intelligence through Large Language Models (LLM) using the Ollama server. By combining game mechanics with intelligent decision analysis, the system provides an engaging and interactive platform for learning entrepreneurship.

The simulation includes several stages of the startup lifecycle such as domain selection, employee hiring, company registration, product development, and product deployment. Each stage represents a game level where the player

must make strategic decisions related to resource allocation, product design, marketing strategies, and financial management.

Through this interactive simulation, users can develop important entrepreneurial skills including:

- Strategic decision making
- Risk analysis
- Resource management
- Business planning
- Market analysis

By simulating real-world startup scenarios, the system provides a practical learning environment where users can experiment with different business strategies without financial risk.

1.2 STARTUP BUSINESS PROCESS SIMULATION

The proposed project "An Intelligent Framework for Modelling and Simulating Startup Business Process" belongs to the interdisciplinary domains of Game Development, Artificial Intelligence, Business Simulation, and Intelligent Decision Support Systems. The system integrates advanced technologies to create an interactive environment where users can simulate the process of building and managing a startup company. The primary objective of the domain is to provide a simulation-based learning platform that enables users to understand the lifecycle of a startup through an engaging and interactive game environment. The developed system uses the Unity game engine for game

development and integrates Artificial Intelligence techniques and Large Language Models (LLM) through the Ollama server to generate intelligent responses, dynamic business scenarios, and real-time feedback.

The application simulates multiple stages of a startup lifecycle, including domain selection, employee hiring, company registration, and product deployment. Each stage acts as a game level where players make strategic decisions related to funding, marketing, product design, and resource management. The system analyzes these decisions using behavioral modeling techniques and evaluates them through a rule-based decision system. By combining game mechanics with AI-based analysis, the platform provides an immersive learning environment that helps players develop entrepreneurial thinking, strategic planning skills, and risk management abilities.

1.2.1 Game Development Domain

The project primarily falls under the Game Development domain, as the system is implemented as an interactive simulation game using the Unity Game Engine. Game development involves designing interactive environments, gameplay mechanics, graphics rendering, and user interface components to create engaging digital experiences. In this project, Unity is used to design the game environment, manage player interactions, and implement the different levels of the startup simulation process. Each phase of the startup lifecycle is represented as a level in the game, allowing players to progress through different business stages while gaining experience points (XP). Game mechanics such as decision-making, scoring systems, and dynamic events are implemented to simulate realistic startup scenarios. The use of game development techniques makes the system interactive, engaging, and suitable for educational purposes.

1.2.2 Artificial Intelligence Domain

The system also belongs to the Artificial Intelligence (AI) domain, as it incorporates intelligent mechanisms to analyze player behavior and generate adaptive gameplay experiences. Artificial Intelligence is used in the system through the integration of Large Language Models (LLM) using the Ollama server. The AI component provides intelligent feedback, generates market analysis, simulates investor responses, and evaluates player decisions. AI-based behavioral modeling techniques are used to analyze the decision-making patterns of players and determine how their choices affect the startup's success. By integrating AI, the system becomes capable of creating dynamic business environments and personalized gameplay scenarios that adapt according to the player's strategies.

1.2.3 Business Simulation Domain

Another important domain of the project is Business Simulation. Business simulation systems are designed to replicate real-world business processes in a controlled virtual environment. In this project, the simulation focuses on the startup ecosystem, allowing players to experience the challenges involved in starting and managing a company. The system simulates business activities such as market analysis, hiring employees, managing budgets, product development, and launching products. The simulation provides players with opportunities to experiment with different strategies and understand the consequences of their decisions without facing real-world financial risks. This approach helps users learn important entrepreneurial concepts and develop practical business management skills.

1.2.4 Intelligent Decision Support Systems

The project also falls under the domain of Intelligent Decision Support Systems (IDSS). Decision support systems assist users in making informed decisions by analyzing available data and providing recommendations. In this system, a rule-based decision engine evaluates player actions by comparing them with predefined optimal startup strategies. The system then generates feedback and suggestions to guide the player toward better decision-making.

1.3 SCOPE OF THE PROJECT

The scope of this project is to create a simulation platform that allows users to understand the process of starting and managing a business. The system focuses on providing an interactive environment where players can explore different strategies for building a successful startup.

The simulation covers multiple stages of the startup lifecycle including business domain selection, company formation, product development, and market deployment. The system evaluates player decisions and provides feedback based on predefined business strategies and artificial intelligence analysis.

The project can be used as an educational tool for students, entrepreneurs, and individuals interested in learning about business management and startup development. It provides a safe virtual environment where users can experiment with different business strategies without facing real financial risks.

Future versions of the system can expand the simulation by introducing additional business sectors, advanced AI capabilities, multiplayer environments, and more complex market dynamics.

CHAPTER 2

LITERATURE SURVEY

2.1 BUSINESS SIMULATION GAMES FOR ENHANCING DECISION-MAKING SKILLS IN HIGHER EDUCATION

Alexander Velez and Rebeca Kerstin Alonso explained that business simulation games are powerful tools used in education to improve decision-making skills and strategic thinking among students. Their research focuses on how simulation-based learning environments can replicate real-world business scenarios and allow learners to practice managerial decision making without real financial risk. The authors emphasize that simulation games provide an interactive platform where users can experiment with different strategies, evaluate outcomes, and learn from their mistakes. According to the authors, traditional learning methods often rely on theoretical explanations and classroom discussions, which may not fully prepare students for real business challenges. Business simulation games overcome this limitation by offering experiential learning opportunities. In these environments, participants act as decision makers who must manage various aspects of a business such as finance, marketing, operations, and human resources. The dynamic nature of the simulation requires players to analyze market conditions, allocate resources effectively, and respond to unexpected challenges. The study also highlights that simulation games promote active learning and engagement. When students participate in simulated business scenarios, they develop critical thinking skills and learn how to evaluate multiple alternatives before making decisions. The interactive nature of the system encourages collaboration, problem-solving, and strategic planning.

2.2 SERIOUS GAMES AS AN EDUCATIONAL TOOL FOR ENTREPRENEURSHIP AND STARTUP TRAINING

Laamarti, Eid, and El Saddik conducted a comprehensive survey on serious games and their applications in various fields, particularly in healthcare and education. The authors describe serious games as digital games designed not only for entertainment but also for educational, training, and problem-solving purposes. Their research highlights the increasing importance of serious games in improving learning outcomes and providing interactive experiences for users. The authors explain that serious games incorporate elements such as simulation, feedback systems, performance tracking, and adaptive learning mechanisms. These features allow users to actively participate in complex scenarios and learn through experience. In many cases, serious games replicate real-world environments where users must make decisions that affect the overall outcome of the system. Another important contribution of the study is the discussion on game design principles for educational purposes. The authors emphasize that effective serious games should combine engaging gameplay with meaningful learning objectives. The system should provide challenges, rewards, and feedback mechanisms that encourage users to continuously improve their skills. The survey also explores the role of emerging technologies such as artificial intelligence, virtual reality, and data analytics in serious game development. These technologies enhance realism and enable systems to analyze player behavior, allowing personalized learning experiences. By evaluating user actions and performance, intelligent systems can adjust difficulty levels and create adaptive scenarios. The authors conclude that serious games have significant potential for improving training and education across multiple domains. The integration of simulation techniques, decision-making tasks, and interactive environments helps users develop practical skills and apply theoretical knowledge in realistic situations.

2.3 GAMIFICATION AND SIMULATION IN ENTREPRENEURSHIP EDUCATION

Pando-Garcia, Periañez-Cañadillas, and Charterina studied the role of gamification and simulation techniques in entrepreneurship education. Their research focuses on how interactive learning environments can improve entrepreneurial skills among students and aspiring business professionals. The authors explain that entrepreneurship education often requires practical understanding of business processes, strategic planning, and decision making. Traditional classroom teaching methods may not fully prepare students for real-world business challenges. Gamified learning systems and simulation games provide an alternative approach by allowing learners to experience realistic business environments. In their study, the authors demonstrate that simulation-based platforms help learners understand concepts such as market competition, financial planning, and product development. Through these systems, participants can test various strategies and analyze the outcomes of their decisions. This experiential learning approach encourages experimentation and improves problem-solving abilities. Another key finding of the research is that gamification elements such as rewards, levels, and achievements increase user engagement and motivation. When learners participate in interactive simulations, they become more involved in the learning process and develop a deeper understanding of business strategies. The authors conclude that integrating simulation games into entrepreneurship education can significantly enhance learning effectiveness. These systems provide practical exposure to business operations and help students develop the skills required for startup management and strategic decision making.

2.4 BUSINESS SIMULATION GAMES IN HIGHER EDUCATION LEARNING

Mustata, Stanescu, and Iacob investigated the effectiveness of business simulation games as learning tools in higher education. Their research focuses on how simulation-based environments can enhance student engagement and improve the understanding of business concepts. The authors highlight that business simulations allow learners to participate in realistic decision-making scenarios where they manage virtual organizations. In these environments, players must analyze market conditions, manage financial resources, and implement strategic decisions to achieve business success. The study also emphasizes that simulation games encourage experiential learning. Instead of passively receiving information, students actively interact with the system and explore different strategies. This hands-on approach allows learners to observe the consequences of their decisions and gain valuable insights into business management. Furthermore, the authors discuss the importance of feedback mechanisms in simulation games. Immediate feedback helps players understand the impact of their actions and refine their strategies. Over time, this process helps learners develop analytical thinking and decision-making abilities. When learners participate in interactive simulations, they become more involved in the learning process and develop a deeper understanding of business strategies. The researchers conclude that business simulation games provide a powerful educational tool for teaching management and entrepreneurship. By combining theoretical knowledge with practical experiences, these systems improve learning effectiveness and prepare students for real-world business challenges.

2.5 GAME-BASED LEARNING FOR STRATEGIC MANAGEMENT EDUCATION

Reilly and Wheatley explored the use of game-based learning systems in strategic management education. Their research highlights how interactive games can help students understand complex management concepts and develop strategic decision-making skills. The authors explain that strategic management involves analyzing competitive environments, identifying opportunities, and making long-term decisions for organizational success. Traditional teaching methods may not provide sufficient practical experience for learners to understand these processes. Game-based learning systems address this challenge by creating simulated business environments where players must make strategic decisions. In these simulations, participants manage virtual companies, develop products, compete with other organizations, and respond to changing market conditions. Another important aspect of the study is the role of feedback and analytics in learning environments. The authors conclude that digital game-based learning platforms have great potential for improving education. These systems combine entertainment and learning to create effective and engaging educational experiences. The authors explain that digital games create interactive learning platforms where students can actively participate in problem-solving activities. Game systems can track player actions and evaluate performance based on predefined metrics. This information allows learners to analyze their strategies and improve their decision-making abilities. The authors conclude that game-based learning provides an effective approach for teaching strategic management concepts. By engaging students in realistic scenarios, these systems enhance critical thinking, problem-solving skills, and strategic planning abilities.

2.6 DIGITAL GAME-BASED LEARNING IN EDUCATIONAL ENVIRONMENTS

Alamri, Watson, and Watson examined the impact of digital game-based learning in modern educational environments. Their research focuses on how interactive gaming systems can enhance student engagement and improve knowledge retention. The authors explain that digital games provide immersive learning experiences where users interact with virtual environments and complete tasks to achieve specific objectives. These systems often include challenges, rewards, and feedback mechanisms that motivate learners to continue exploring the content. The study also highlights the importance of adaptive learning systems in digital games. By analyzing user performance and behavior, the system can adjust the difficulty level and provide personalized learning experiences. The researchers found that students who participate in game-based learning environments demonstrate higher levels of engagement and motivation compared to traditional learning methods. Interactive systems encourage exploration and experimentation, allowing learners to develop practical skills. The authors conclude that digital game-based learning platforms have great potential for improving education. These systems combine entertainment and learning to create effective and engaging educational experiences. The authors explain that digital games create interactive learning platforms where students can actively participate in problem-solving activities. These systems use challenges, rewards, and feedback mechanisms to maintain learner motivation and engagement. The study highlights that adaptive learning systems can analyze player behavior and adjust the difficulty level accordingly. This helps provide personalized learning experiences for different users. The researchers found that students who learn through game-based environments show higher engagement and better knowledge retention. The study concludes that digital game-based learning improves motivation, critical thinking, and learning outcomes.

2.7 EXPERIENTIAL LEARNING THROUGH BUSINESS SIMULATION GAMES

Ferreira, Cardoso, and Abrantes studied the role of business simulation games in experiential learning. Their research highlights how simulation environments allow learners to develop practical skills by engaging in realistic business scenarios. The authors explain that experiential learning involves learning through direct experience rather than theoretical instruction. Business simulation games support this approach by allowing users to manage virtual companies and make decisions that influence the outcome of the system. In these environments, players must analyze market trends, allocate resources, and implement strategies to achieve business success. By observing the results of their actions, learners gain insights into complex business processes. The study also emphasizes the role of teamwork and collaboration in simulation games. Many business simulations require participants to work together, share information, and coordinate strategies to achieve common goals. The authors conclude that business simulation games provide an effective platform for experiential learning. These systems help learners develop decision-making skills, strategic thinking abilities, and a deeper understanding of business operations. The authors state that simulation games allow students to gain practical experience by managing virtual companies. In these environments, players make decisions related to finance, marketing, and resource management. The system provides feedback based on the player's decisions, helping them understand the impact of their strategies. Simulation platforms also encourage teamwork and collaborative problem solving among participants. Through repeated experimentation, learners develop better strategic thinking abilities. The study concludes that simulation-based learning effectively bridges the gap between theoretical knowledge and practical business experience.

2.8 SIMULATION BASED LEARNING FOR ENTREPRENEURSHIP DEVELOPMENT

Mannino, Nania, and Fichera examined the use of simulation-based learning systems for entrepreneurship development. Their research focuses on how virtual business environments can support the training of future entrepreneurs. The authors explain that entrepreneurship requires the ability to analyze opportunities, manage risks, and develop innovative solutions. Simulation platforms allow learners to practice these skills in a controlled environment where mistakes do not result in real financial losses. Through simulation systems, users can create virtual startups, develop products, and compete in dynamic markets. The system tracks player decisions and evaluates performance based on factors such as profitability, market share, and innovation. The study also highlights the benefits of integrating artificial intelligence into simulation environments. AI systems can analyze user behavior, provide recommendations, and create adaptive scenarios that challenge players. The authors conclude that simulation-based learning platforms are valuable tools for entrepreneurship education. These systems provide practical training opportunities and help learners develop the skills required for startup success. The authors explain that entrepreneurship requires skills such as innovation, decision making, and risk management. Simulation platforms provide virtual environments where learners can create and manage startup companies. These systems allow players to experiment with different business strategies without real financial risk. By analyzing the results of their decisions, users gain insights into real-world business operations. The study also highlights the use of intelligent systems to track player behavior and performance. The research concludes that simulation-based learning helps students develop entrepreneurial thinking and strategic planning skills.

2.9 GAME PLATFORMS FOR DECISION-MAKING SKILL DEVELOPMENT

Grijalvo, García-Wunsch, and López studied the use of game-based platforms for developing decision-making skills. Their research highlights how interactive simulations can improve analytical thinking and strategic planning abilities. The authors explain that decision-making is a critical skill in many professional fields, including business management and entrepreneurship. Simulation games provide realistic scenarios where users must analyze information and choose appropriate strategies. In these systems, players receive feedback based on their decisions. This feedback helps them understand the consequences of their actions and refine their strategies over time. By repeating the decision-making process in different scenarios, users gradually improve their skills. The study also discusses the importance of data analytics in evaluating player performance. By analyzing gameplay data, researchers can identify behavioral patterns and measure learning outcomes. The authors conclude that game-based platforms are effective tools for training decision-making skills. These systems provide interactive environments where learners can practice strategic thinking and problem-solving. The authors explain that interactive simulation games allow players to analyze different scenarios before making strategic decisions. These systems often include dynamic environments where market conditions and resources continuously change. Players must evaluate available information and select the best possible strategies. The feedback provided by the system helps learners understand the consequences of their decisions. Over time, users develop better analytical thinking and problem-solving abilities. The study concludes that simulation games are effective tools for developing decision-making skills in educational environments.

2.10 ARTIFICIAL INTELLIGENCE IN GAME-BASED LEARNING SYSTEMS

Huang, Hew, and Lo investigated the integration of artificial intelligence technologies in game-based learning systems. Their research focuses on how AI can enhance educational games by providing intelligent feedback and personalized learning experiences. The authors explain that AI technologies such as machine learning, natural language processing, and behavioral analytics can analyze player interactions and adapt the game environment accordingly. This allows systems to create dynamic learning experiences tailored to individual users. The study also highlights the use of intelligent tutoring systems within games. These systems provide guidance and recommendations based on player performance, helping learners improve their strategies and understanding of the subject. Another important contribution of the research is the discussion on adaptive gameplay mechanisms. AI systems can modify game scenarios, adjust difficulty levels, and introduce new challenges based on player behavior. The authors conclude that the integration of artificial intelligence significantly enhances the effectiveness of game-based learning systems. By providing personalized feedback and adaptive gameplay, AI-driven platforms can improve learning outcomes and user engagement. The authors explain that AI technologies such as machine learning and behavioral analytics help analyze player interactions within games. These systems can provide personalized feedback and recommendations based on user performance. AI-driven mechanisms also enable adaptive gameplay where difficulty levels change according to the player's skill level. This improves user engagement and learning effectiveness.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

Existing systems related to business education and entrepreneurship training mainly rely on traditional teaching methods and basic simulation tools. In many educational institutions, entrepreneurship and business management concepts are taught through lectures, case studies, and theoretical discussions. While these methods provide fundamental knowledge about business operations, they often lack practical exposure and real-world experience for students. As a result, learners may find it difficult to understand how business strategies and decision-making processes work in dynamic market environments. Some existing platforms use business simulation software to demonstrate how companies operate in competitive markets. These systems allow users to manage virtual businesses by making decisions related to production, marketing, finance, and resource allocation. However, many of these systems follow static gameplay models where the scenarios are predefined and do not change based on player behavior. This limits the ability of the system to provide personalized learning experiences.

In addition, most existing simulation platforms do not incorporate advanced artificial intelligence techniques to analyze user behavior or decision-making patterns. They generally provide limited feedback to players and do not adapt the game environment according to user performance. As a result, the learning experience may become repetitive and less engaging for players. Another limitation of many existing systems is the lack of integration between modern technologies such as game engines, AI-based analytics, and intelligent decision-support systems. Traditional business simulation tools often focus only

on numerical calculations or simple strategy models without providing immersive interactive environments.

Furthermore, existing platforms rarely simulate the complete lifecycle of a startup company. Most systems focus on individual aspects such as financial planning or market competition but do not cover the entire process of building a startup from idea generation to product deployment. These limitations highlight the need for more advanced systems that integrate artificial intelligence, adaptive gameplay, and interactive game development platforms to create a more realistic and engaging startup simulation experience.

3.2 BLOCK DIAGRAM OF EXISTING SYSTEM

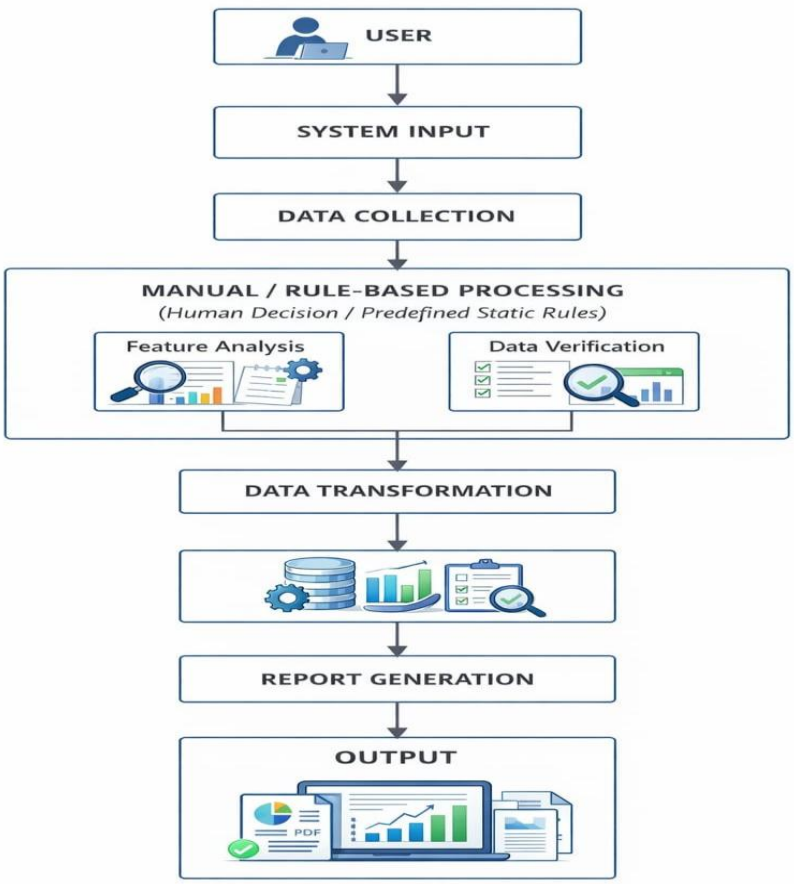

Figure. 3.1 Block diagram of existing system

CHAPTER 4

PROBLEMS IDENTIFIED

Lack of Practical Learning Environment

Most traditional business education systems rely heavily on theoretical concepts and classroom lectures. Students learn business strategies from textbooks without experiencing real-world decision-making situations. This creates a gap between theoretical knowledge and practical application. Learners often struggle to understand how business strategies work in dynamic environments. Without practical exposure, students may find it difficult to apply their knowledge in real startup scenarios. Therefore, the absence of an interactive learning environment limits the effectiveness of business education.

Absence of Intelligent Decision Analysis

Existing systems generally do not analyze the decisions made by users during gameplay. They only display results based on predefined calculations without evaluating the strategy behind the decisions. Decision analysis is important for understanding user behavior and improving strategic thinking. Without intelligent analysis, the system cannot provide meaningful feedback to players. This reduces the educational value of the simulation. Hence, intelligent decision evaluation is necessary for effective learning.

Lack of Adaptive Gameplay Mechanisms

Most simulation systems follow fixed scenarios that remain the same for all users. These static environments do not change according to the player's performance or learning progress. Adaptive gameplay is important for maintaining engagement and improving learning outcomes. Without adaptive

mechanisms, the game may become repetitive and less challenging. Players may lose interest if the system does not respond to their actions. Therefore, dynamic and adaptive gameplay features are required.

Inadequate Feedback and Guidance

Many existing platforms provide only basic feedback to users after completing tasks. They do not offer detailed explanations about why certain decisions are correct or incorrect. Effective learning requires continuous feedback and guidance. Without proper feedback, players may repeat the same mistakes without understanding their impact. This reduces the educational value of the system. Therefore, improved feedback mechanisms are necessary.

Limited Coverage of Startup Lifecycle

Most current business simulation systems focus only on specific aspects such as financial planning or marketing strategies. They do not simulate the complete lifecycle of a startup business. Starting a company involves multiple stages such as idea generation, domain selection, company registration, and product deployment. Without covering all these stages, learners cannot understand the entire startup development process. Therefore, a comprehensive simulation system is needed.

Lack of Integration with Modern Technologies

Many traditional simulation platforms do not integrate modern technologies such as artificial intelligence, game engines, or machine learning systems. These technologies can significantly improve the realism and effectiveness of learning environments. Without them, the system may remain limited in functionality and intelligence. Modern technologies enable adaptive learning and advanced analytics. Therefore, integrating advanced technologies is necessary for improving simulation systems.

Low User Engagement and Motivation

In many traditional learning platforms, the lack of interactive elements reduces user engagement. Players may feel bored when the system only presents static information or repetitive tasks. Engagement is an important factor in maintaining user interest and improving learning outcomes. Game-based systems can provide motivation through levels, rewards, and achievements. Without such elements, users may lose interest quickly. Hence, improving engagement is essential.

Absence of AI-Based Personalized Learning

Existing systems rarely use artificial intelligence to personalize the learning experience. Each user receives the same content and scenarios regardless of their skill level or decision patterns. Personalized learning can help users improve by providing tailored challenges and feedback. AI technologies can analyze player behavior and adjust the system accordingly. Without personalization, the learning experience may not be effective for all users. Therefore, AI-based personalization is needed to enhance learning outcomes.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM

The proposed system is an AI-driven startup simulation game designed to provide an interactive and practical learning environment for understanding the process of building and managing a startup company. The system is developed using the Unity game engine, which allows the creation of immersive gameplay environments and interactive user interfaces. The main objective of the proposed system is to simulate real-world business scenarios where players can experience the challenges of running a startup and making strategic decisions. In the proposed system, the player progresses through a structured four-phase startup development process, where each phase is represented as a game level. These phases include Domain Selection, Company Registration, Product Development, and Product Deployment. In the first phase, players select a business domain such as technology, healthcare, or e-commerce based on market demand and available resources. The second phase involves the process of registering the company, choosing the company structure, and completing legal requirements. In the third phase, players focus on product development, where they allocate resources, manage employees, and design products based on market needs. Finally, in the product deployment phase, the player launches the product in the market and implements marketing and sales strategies to achieve business growth. A key feature of the proposed system is the integration of Artificial Intelligence and Large Language Models (LLMs) using the Ollama server. This AI component analyzes player decisions and generates intelligent feedback based on the actions taken during gameplay. The system uses behavioral modeling techniques to study the player's decision-making patterns and evaluate how effectively they manage resources and respond to business challenges. The system also incorporates a rule-based

decision evaluation mechanism, which compares the player's decisions with predefined optimal business strategies. Based on this comparison, the system provides suggestions and guidance to help players improve their strategic thinking skills. Additionally, the game includes an adaptive gameplay mechanism, where the difficulty level, market competition, and financial risks dynamically change based on the player’s performance.

5.2 BLOCK DIAGRAM OF PROPOSED SYSTEM

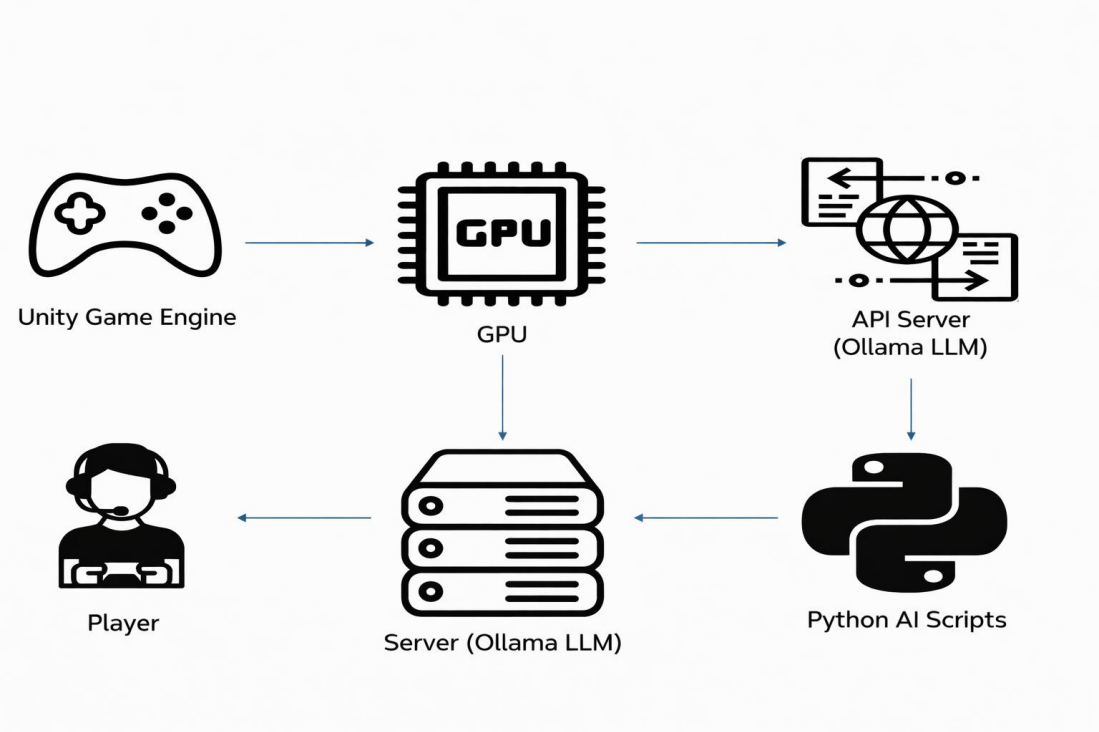

Figure. 5.1 Block Diagram of Proposed System

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5
RAM	Minimum 8 GB
Storage	256 GB SSD or higher
GPU	NVIDIA RTX 2050

6.2 SOFTWARE REQUIREMENTS

Component	Specification
Operating System	Windows 10/11
Game Engine	Unity Game Engine
Programming Language	C# for Unity scripting
AI Framework	Ollama LLM Server for AI-based decision analysis
Development Tools	Visual Studio Code for coding and debugging

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The proposed system is divided into several functional modules to ensure efficient development and smooth operation of the startup simulation game. Each module is responsible for performing a specific task within the overall system. These modules work together to simulate the different stages involved in building and managing a startup company. By dividing the system into modules, the development process becomes more organized and easier to manage. The modular structure also allows individual components to be modified or improved without affecting the entire system. This approach improves system maintainability, scalability, and performance.

7.2 MODULES DESCRIPTION

7.2.1 Player Interaction Module

7.2.2 Domain Selection Module

7.2.3 Company Registration Module

7.2.4 Product Development Module

7.2.5 Product Deployment Module

7.2.1 Player Interaction Module

The Player Interaction Module manages all interactions between the player and the game environment. It provides the user interface through which players can navigate different game levels and make decisions. This module captures player inputs such as selections, commands, and actions performed during gameplay. It ensures that user actions are correctly

interpreted and executed by the system. The module also displays game information such as resources, progress, and feedback messages. By providing a smooth and interactive interface, this module enhances the overall gaming experience. It acts as the communication layer between the player and the game system. The module ensures that gameplay remains intuitive and user-friendly.

7.2.2 Domain Selection Module

The Domain Selection Module represents the first stage of the startup simulation game. In this module, players choose the business domain in which they want to establish their startup. Examples of domains include technology, healthcare, e-commerce, and education. The system provides information about each domain, including market demand, competition, and potential opportunities. Based on this information, the player selects a suitable domain for their business idea. The chosen domain influences the challenges and opportunities presented in later stages of the game. This module encourages players to analyze market conditions before making decisions. It helps players understand the importance of selecting the right business sector. The domain selection process plays a crucial role in shaping the overall gameplay experience.

7.2.3 Company Registration Module

The Company Registration Module simulates the process of establishing a new business entity. In this stage, the player selects the company name, business structure, and basic organizational details. The system guides the player through the steps required for company registration. This module introduces players to the legal and administrative aspects of starting a business. It also requires players to allocate initial resources and complete necessary setup activities. By simulating these processes, the system provides a realistic understanding of startup formation. The module helps players recognize the

importance of proper planning and legal compliance. This stage prepares the player for the operational aspects of running the business.

7.2.4 Product Development Module

The Product Development Module focuses on creating and designing the product or service offered by the startup. In this stage, players decide how to allocate resources such as budget, employees, and time for product creation. The system provides options for selecting product features, quality levels, and development strategies. Players must consider factors such as customer needs, market demand, and production cost. The decisions made during this stage influence the success of the product in the market. The module helps players understand the importance of innovation and effective resource management. It also demonstrates how product quality and development planning affect business outcomes. This stage prepares the startup for entering the competitive market.

7.2.5 Product Deployment Module

The Product Deployment Module represents the final stage of the startup simulation process. In this module, players launch their product into the market and implement marketing strategies. The system simulates real-world market conditions such as competition, customer demand, and financial risks. Players must make decisions related to pricing, advertising, and distribution. The success of the product depends on the strategies chosen by the player. The system analyzes player decisions and provides feedback based on performance. This module helps players understand how business strategies affect market success. It completes the startup lifecycle simulation by demonstrating the challenges involved in product launch and market growth.

CHAPTER 8

SYSTEM TESTING

8.1. SYSTEM TESTING

System testing is an important phase in the software development process that ensures the developed system functions correctly and meets the specified requirements. It involves evaluating the complete and integrated software system to verify that all modules work together as expected. The main objective of system testing is to identify errors, bugs, or performance issues before the system is deployed for actual use.

Testing also verifies that user interactions, decision analysis, and adaptive gameplay mechanisms function properly. Through system testing, developers can validate the system's reliability, performance, and usability. The testing process includes multiple types of testing such as Unit Testing, Integration Testing, System Testing, Performance Testing, Security Testing, and Usability Testing. Each type of testing focuses on a specific aspect of the system to ensure overall quality and functionality.

8.2. UNIT TESTING

Unit testing is the process of testing individual components or modules of a software system to ensure they function correctly. Each module is tested separately to verify that it performs its intended task without errors. In the proposed system, unit testing is performed on modules such as domain selection, company registration, and product development.

During unit testing, developers check whether each module processes inputs correctly and produces the expected outputs. For example, the domain

selection module is tested to ensure that the selected domain is properly stored and displayed in the game interface. Unit testing helps identify errors at an early stage and ensures that individual components work correctly before integration with other modules.

8.3. INTEGRATION TESTING

Integration testing focuses on verifying the interaction between different modules of the system. After individual modules are tested, they are combined and tested as a group to ensure they work together correctly. This testing ensures that data flows properly between modules and that the system functions as an integrated whole.

In the startup simulation game, integration testing is used to verify the connection between the gameplay modules and the AI decision analysis system. For example, when a player selects a domain and proceeds to company registration, the system must correctly transfer the selected data to the next stage. Integration testing ensures that transitions between game levels and modules operate smoothly without errors.

8.4. PERFORMANCE TESTING

Performance testing evaluates how well the system performs under different conditions such as heavy workloads or extended gameplay sessions. The goal is to ensure that the system remains stable and responsive during operation.

For the startup simulation game, performance testing checks factors such as game loading time, response time for player actions, and smooth execution of gameplay scenarios. The testing process ensures that the Unity game engine runs efficiently and that the integration with AI components does not cause

delays or performance issues. Proper performance testing ensures that users experience smooth and uninterrupted gameplay.

8.5 SECURITY TESTING

Security testing is performed to ensure that the proposed startup simulation game is protected from unauthorized access, data manipulation, and system misuse. The main objective of security testing is to verify that the system maintains the confidentiality, integrity, and safety of the data exchanged between different components. Security testing further ensures that the application does not crash or behave abnormally when incorrect or unusual inputs are given by the user.

It helps in identifying possible vulnerabilities in input handling, data exchange, and system execution. By performing security testing, the reliability and trustworthiness of the startup simulation game are improved. Therefore, this testing is essential for maintaining a stable and secure gameplay environment.

8.6 USABILITY TESTING

Usability testing is conducted to evaluate how easily players can interact with the startup simulation game and understand its features. The main purpose of usability testing is to ensure that the system is user-friendly, simple to navigate, and provides a smooth gaming experience. The clarity of instructions, menu design, navigation controls, and overall interface layout are tested to determine whether the user can operate the system efficiently.

Special attention is given to ensuring that players can clearly understand the game objectives and receive feedback in an understandable format.

CHAPTER 9

RESULTS AND DISCUSSION

The proposed system, an Intelligent Framework for Modelling and Simulating Startup Business Processes, was successfully developed using the Unity game engine integrated with artificial intelligence components. The system was designed to simulate the process of building and managing a startup company through an interactive game-based learning environment. The game includes multiple stages such as domain selection, company registration, product development, and product deployment. Each stage represents a level in the game where players make strategic decisions that affect the progress and success of the virtual startup. During the implementation phase, the system was tested to ensure that all modules functioned properly and interacted seamlessly with one another. The results show that the game successfully provides a realistic simulation of startup development processes. Players were able to navigate through the different levels of the game and make decisions related to business strategies, resource management, and product development. The Unity game engine provided a dynamic and visually interactive environment that enhanced user engagement and made the learning experience more effective. One of the key features of the developed system is the integration of Artificial Intelligence using Large Language Models (LLM) through the Ollama server.

The AI component analyzes the decisions made by the player and generates feedback based on predefined business rules and behavioral patterns. This intelligent feedback mechanism helps players understand the impact of their decisions and encourages them to improve their strategic thinking skills. The system successfully demonstrated the ability to evaluate player actions and provide suggestions that guide users toward better decision-making strategies. The results also indicate that the rule-based decision system effectively

compares player choices with optimal business strategies stored in the system. When players make decisions that do not align with recommended strategies, the system provides informative feedback explaining the potential consequences. This feature helps users learn from their mistakes and understand how different decisions influence the success of a startup. By providing continuous evaluation and guidance, the system enhances the educational value of the simulation. Another important aspect observed during testing was the effectiveness of the adaptive gameplay mechanism. The game environment dynamically adjusts various factors such as market demand, competition level, and financial risk based on the player's performance. For example, when players successfully complete certain stages, the game introduces more challenging scenarios to test their strategic abilities. This adaptive feature ensures that the gameplay remains engaging and challenging while also supporting personalized learning experiences.

User testing also showed that the system provides an intuitive and user-friendly interface. Players were able to easily understand the game mechanics and navigate through the different stages without difficulty. The graphical interface developed using Unity allows players to visualize the progress of their startup and monitor key business metrics such as financial resources, product development status, and market performance. The interactive design of the system contributes to a more engaging and immersive gameplay experience. In addition to educational benefits, the system also demonstrates the potential of combining game development technologies with artificial intelligence to create advanced simulation platforms. The integration of Unity, AI-based decision analysis, and adaptive gameplay mechanisms creates a powerful environment for learning and experimentation. Such systems can be used not only for educational purposes but also for entrepreneurship training and professional skill development. Overall, the developed startup simulation game successfully achieved its primary objectives. The system effectively demonstrates how interactive gaming environments can be used to simulate real-world business

processes and improve decision-making skills. The integration of AI technologies enhances the system's ability to analyze player behavior and provide meaningful feedback. Through this project, it is evident that intelligent simulation games can serve as valuable tools for teaching entrepreneurship and business management concepts in an engaging and practical manner. The results obtained from the system testing and user interaction confirm that the proposed framework is capable of providing a realistic and effective startup simulation environment. The system allows players to experience the challenges of running a business while learning important concepts such as strategic planning, resource allocation, and market analysis. Therefore, the developed system can be considered a useful platform for both educational and training purposes in the field of entrepreneurship and business management.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The proposed project, “An Intelligent Framework for Modelling and Simulating Startup Business Processes,” was successfully designed and developed as an interactive startup simulation game. The main objective of the project was to create a learning platform that allows users to understand the process of building and managing a startup company through a game-based environment. By integrating modern technologies such as the Unity game engine, artificial intelligence, and large language models, the system provides a practical and engaging approach to learning entrepreneurship and business management concepts. The developed system simulates the different stages involved in establishing a startup company. These stages include domain selection, company registration, product development, and product deployment, which are represented as levels within the game. Each level allows the player to make important decisions related to business strategy, resource management, and market planning. The interactive gameplay environment encourages players to analyze different situations and select appropriate actions that influence the overall success of the startup. One of the key strengths of the proposed system is the integration of AI-based decision analysis using the Ollama LLM server. The artificial intelligence component evaluates player decisions and provides intelligent feedback based on predefined business rules and behavioral patterns. This feature helps players understand the impact of their actions and improves their strategic thinking and decision-making abilities. The rule-based decision system also compares player choices with optimal business strategies, allowing users to learn from their mistakes and

make better decisions in future scenarios. Another important feature of the system is the adaptive gameplay mechanism, which dynamically adjusts game scenarios such as market demand, competition, and financial challenges based on player performance. This ensures that the game remains engaging and challenging for users with different skill levels. The use of the Unity game engine also contributes to an interactive and visually appealing user interface, making the learning experience more immersive and enjoyable. Overall, the project successfully demonstrates how game development and artificial intelligence technologies can be combined to create an effective simulation platform for entrepreneurship education. The system provides users with a safe environment where they can experiment with different business strategies and gain practical insights into startup management. Therefore, the developed startup simulation game can serve as a valuable educational tool for students, entrepreneurs, and individuals interested in learning about business processes and decision making.

10.2 FUTURE ENHANCEMENTS

Although the proposed system successfully simulates the startup development process, several improvements and enhancements can be implemented in the future to further increase its functionality and effectiveness. Future development can focus on expanding the game environment, improving artificial intelligence capabilities, and enhancing the overall user experience. One possible enhancement is the addition of more business domains and industry sectors within the game. Currently, the system includes a limited number of domains for startup creation. In the future, more sectors such as finance, agriculture, artificial intelligence, and digital services can be introduced. This would allow players to explore a wider range of business opportunities and understand how strategies vary across different industries. Another improvement could involve implementing multiplayer functionality,

where multiple users can interact within the same simulation environment. Players could compete with each other by managing their own startup companies in a shared virtual market. This feature would introduce realistic competition and collaboration, making the simulation more engaging and reflective of real-world business environments. The system can also be enhanced by integrating advanced artificial intelligence techniques such as machine learning and predictive analytics. These technologies can be used to analyze player behavior more accurately and provide more personalized feedback. The AI system could also generate dynamic market scenarios, simulate economic fluctuations, and introduce unexpected business challenges to make the simulation more realistic. Future versions of the system may also include improved graphical interfaces and immersive technologies such as virtual reality (VR) or augmented reality (AR). These technologies can create more immersive gameplay experiences and allow users to interact with the virtual business environment in a more natural way. Additionally, the platform could be expanded into a comprehensive educational tool for entrepreneurship training programs. Educational institutions and training organizations could use the system to teach business concepts through interactive simulations. Features such as progress tracking, performance analysis, and instructor dashboards could be added to support classroom learning environments. In conclusion, the proposed startup simulation game provides a strong foundation for developing intelligent learning platforms that combine game technology, artificial intelligence, and business simulation. With further enhancements and technological advancements, the system can evolve into a powerful tool for entrepreneurship education, strategic decision training, and business skill development.

APPENDIX – A
SOURCE CODE

Movement,c#

```
using UnityEngine;

using TMPro;

using UnityEngine.UI;

public class Movement : MonoBehaviour

{

    float xRotation = 0f;

    float yRotation = 0f;

    public float sensitivity = 10f;

    Rigidbody rb;

    public GameObject bicycle;

    public GameObject sit;

    Movement m;

    public InCyclePlayer mPlayer;

    public Camera cam;

    public Camera cam2;
```

```
Ray r;

RaycastHit hit;

public LayerMask layeer;

public TextMeshPro text;

public LayerMask layer2;

public GameObject EntireUi;

public int intract = 1;

public GameObject worstagram;

public GameObject job;

public GameObject pannel;

int complete = 0;

public TextMeshProUGUI t;

public Button btn;

public static string name;

public GameObject Registet;

public TextMeshProUGUI REGISETER;

public TextMeshPro nameOfTheShop;

public Button btns;
```

```
private void Start()
{
    rb = GetComponent<Rigidbody>();
    m= GetComponent<Movement>();
    mPlayer.enabled = false;
    cam.enabled = false;
    pannel.SetActive(false);
    Registet.SetActive(!true);
    btn.onClick.AddListener(KeepName);
    btns.onClick.AddListener(reg);
}
void Update()
{
    float mousePos= Input.GetAxis("Mouse Y") * sensitivity;
    r = Camera.main.ScreenPointToRay(transform.position);
    xRotation -= mousePos;
   // Debug.Log(xRotation);
    xRotation = Mathf.Clamp(xRotation, -37.04f, 37.04f);
```

```
float mousePos1 = Input.GetAxis("Mouse X") * sensitivit

yRotation -= -mousePos1;

//yRotation -= -mousePos1;

transform.rotation = Quaternion.Euler(xRotation, yRotation, 0f);

if(Input.GetKey(KeyCode.W)==true)

{

   rb.velocity = new Vector3(transform.forward.x, 0, transform.forward.z)*5;

}

if(Input.GetKey(KeyCode.W) == false)

{

   rb.velocity = new Vector3(transform.forward.x, 0, transform.forward.z)*0;

}

if(Vector3.Distance(transform.position,bicycle.transform.position)<=3)

{

   if(Input.GetKeyDown(KeyCode.F))

   {

      m.enabled = false;

      cam.enabled = true;
```

```
        mPlayer.enabled = true;
        cam2.enabled = false;
    }
}
if (Physics.Raycast(r, out hit, 100, layeer))
{
    text.text = "Press E to interact";
    if(Input.GetKeyDown(KeyCode.E))
    {
        Debug.Log("ok");
    }
}
if(Physics.Raycast(r,out hit,200,layer2))
{
   if(Input.GetKeyDown(KeyCode.E))
    {
        worstagram.SetActive(true);
        job.SetActive(false);
```

```
            EntireUi.SetActive(true);
            intract = 0;
        }
    }
    else
    {
        //text.text = "Come near";
    }
    if(Input.GetKeyDown(KeyCode.Escape))
    {
        pannel.SetActive(false);
        Registet.SetActive(false);
    }
    if(complete==2)
    {
        nameOfTheShop.text = name;
    }
}
```

```
private void OnTriggerStay(Collider other)
{
    if(other.tag=="npc")
    {
        Debug.Log("notu");
    }
    if (other.tag == "name")
    {
        if(Input.GetKeyDown(KeyCode.E)&&complete==0)
        {
            pannel.SetActive(true);
        }
    }
    if(other.tag=="register")
    {
        if (Input.GetKeyDown(KeyCode.E) && complete == 1)
        {
            REGISETER.text = "comany Name: "+name;
```

```
            Registet.SetActive(true);
        }
    }
}
void KeepName()
{
    name = t.text;
    Debug.Log(name);
    complete = 1;
}
void reg()
{
    complete = 2;
    Registet.SetActive(false);
}
}
```

APPENDIX – B

SCREENSHOTS

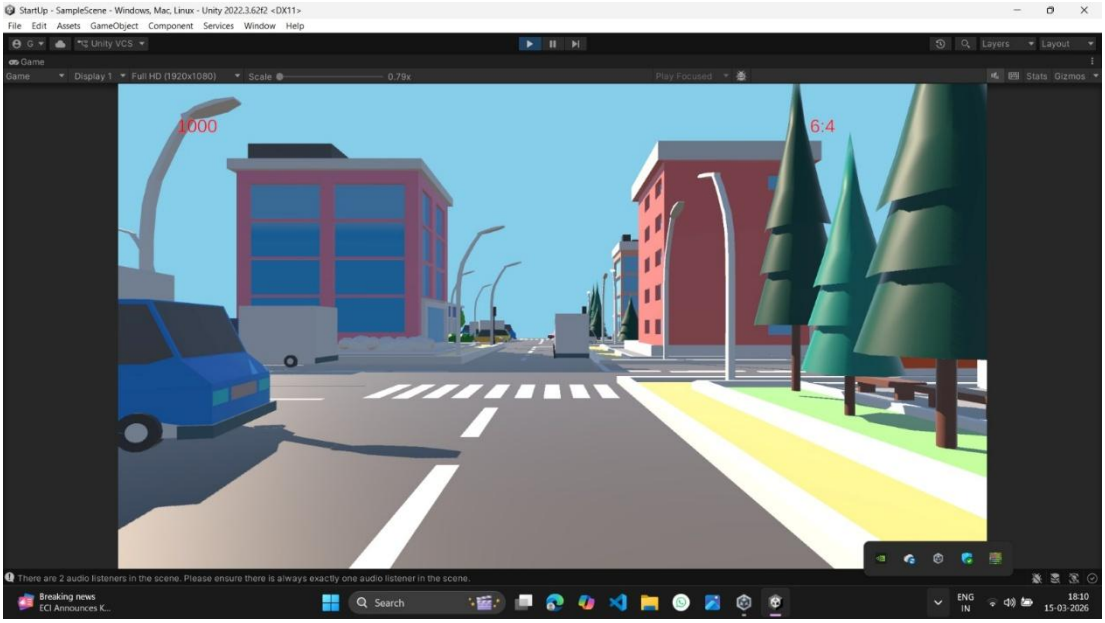

Figure B.1 Game Map

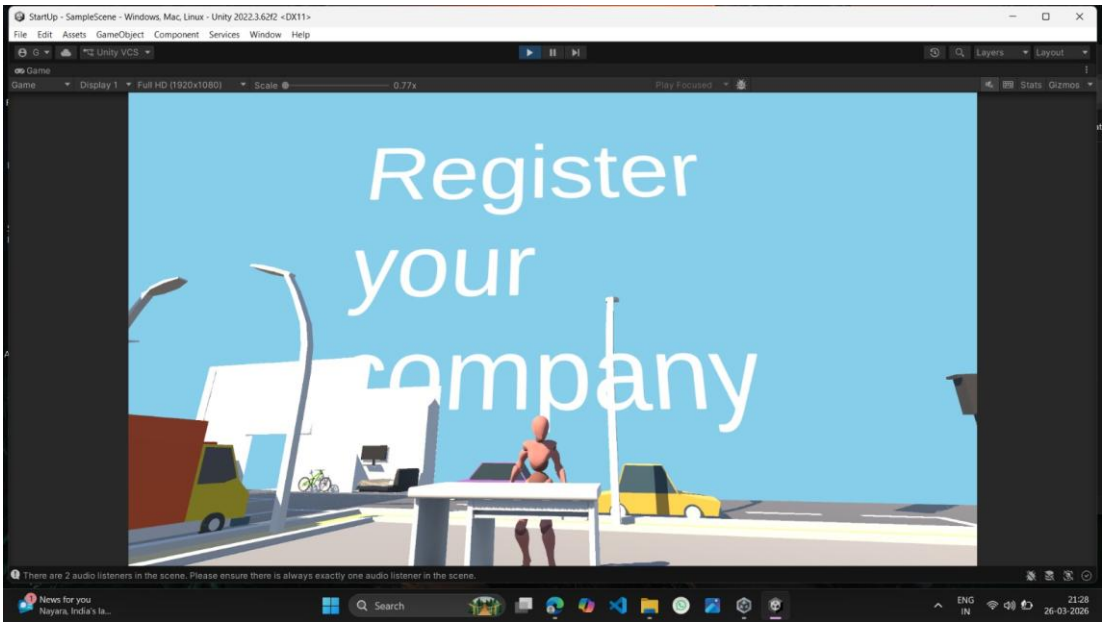

Figure B.2 Registration section

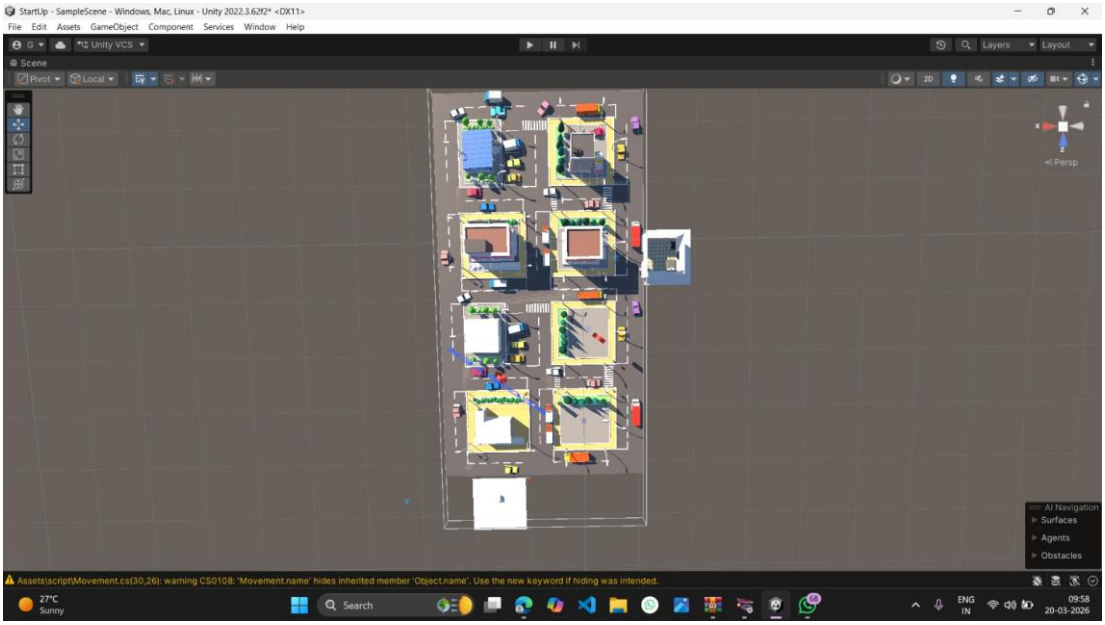

Figure B.3 3d map

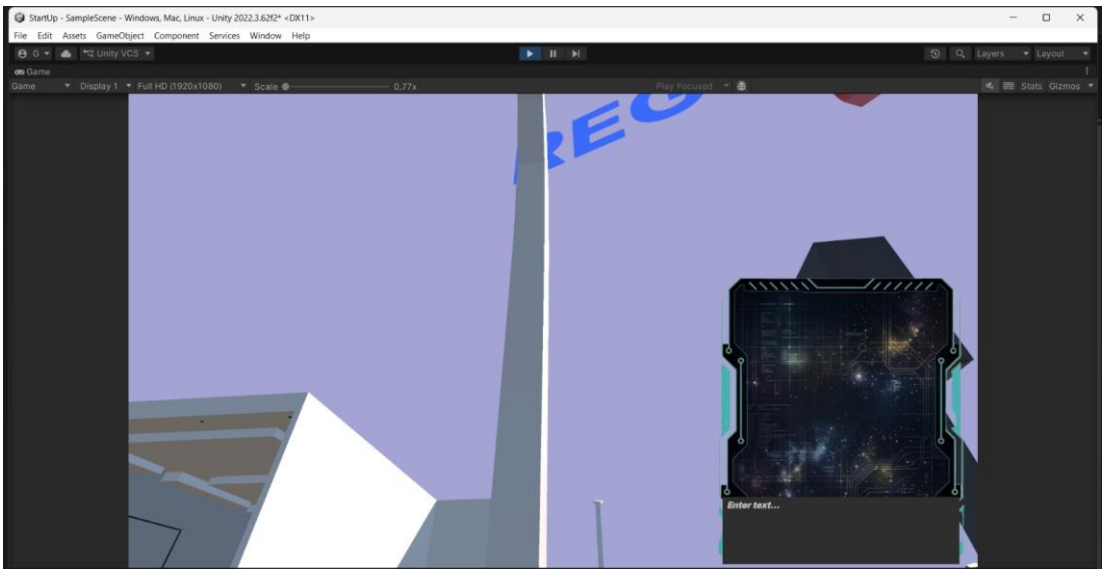

Figure B.4 Chat bot interface

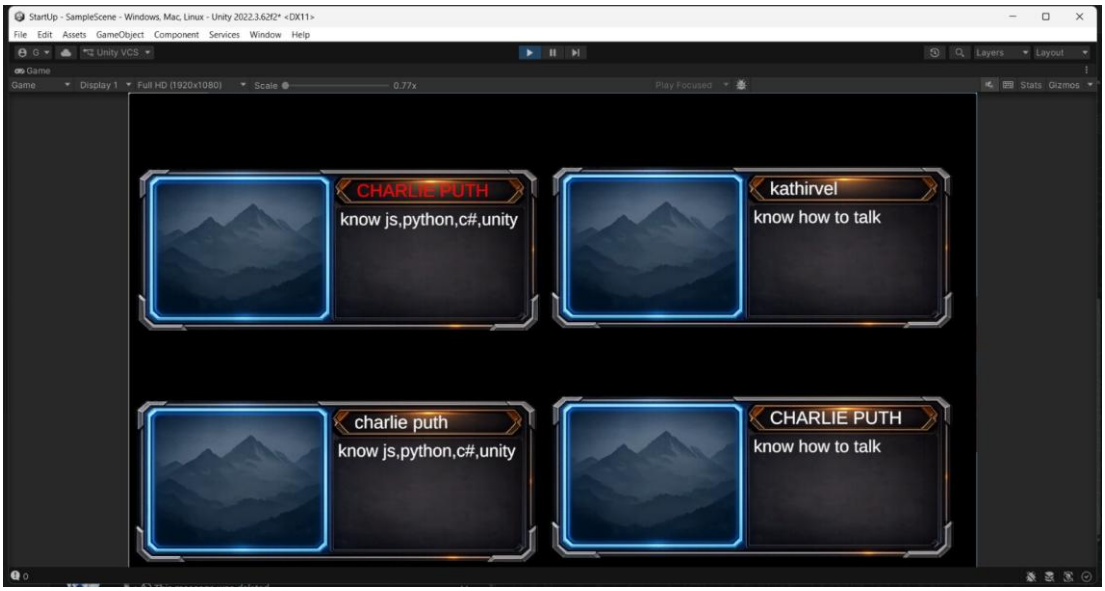

Figure B.5 Hiring Section

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